

CQE_CMPLX

Master PDF

A Folded Strand from Enumerated Parts

Closed-Form Algebra Edition

The Single Observation:

A self-complementary strand folded from 144 enumerated parts through 33

operations yields a stable folded form in which one hydrogen bond reads identically from both strands; this single observation retroactively certifies the entire 33-step folding pathway.

CQE_CMPLX Corpus

33 papers, 144 tools, 32 theorems, 8 colors, 12 tool classes, 1 observation

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End of Closed-Form Algebra

Part 1: Closed-Form Algebra

This document presents the CQE_CMPLX corpus as a complete closed-form

colors, 12 tool classes, and 1 observation - all unified into a single

The algebra has 7 sections:

II. The Color Algebra (8 colors, S4 symmetry)

IV. The Theorem Algebra (32 axioms, 4 derivation rules)

VI. The Bilateral Algebra (2-category)

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Section I. The Gluon Algebra

I.1 Definition. A Gluon is a typed quantity C in the set of bit-strings with:

(b) A receipt (input, T , output, O) (replayable record)

I.2 The 5 Primitive Operations.

Op-1: Side-Flip ($T_BIJECTIVE$)

Type: 1 Gluon $\rightarrow$ 1 Gluon

Fixed point: $(1, 0, 1)$ is the unique non-trivial fixed point

Op-2: Correction ($T_CORRECTION$)

Type: 1 Gluon $\rightarrow$ 1 bit

Fires at: D4 axes $(2, 0)$ and $(3, 1)$ - 3 of 8 states

Op-3: Triality ($T_TRIALITY$)

Type: 1 Gluon $\rightarrow$ 1 Gluon

Orbits: 2 fixed points (vacua) + 2 orbits of 3 (excited) = 8 states

Op-4: Oloid Midpoint ($T_BOUNDARY_REPAIR$)

Type: 2 Gluons $\rightarrow$ 1 Gluon

Existence: unique

Op-5: Accumulate (T_OLOID_PATH)

Type: n Gluons $\rightarrow$ 1 Gluon

Invariant: independent of path

I.3 The 5 Algebraic Identities. The 5 operations satisfy:

Id-2: $\text{corr}(C) = 0$ iff C in $\{(0,0,0),(1,0,0),(0,0,1),(1,0,1),(0,1,1)\}$ (5

Id-3: $\text{tri}^3 = \text{id}$ (triality is order 3)

midpoint)

Theorem I.4.1 (Gluon algebra is closed). The 5 operations form a closed

Proof: From the verification of $T_BIJECTIVE$, $T_CORRECTION$, $T_TRIALITY$,

state space. The compositions are well-defined by the algebraic identities of

(a) A res

(c) An i c

$\text{sf} : (L,$

Idempot en

$\text{corr} : (I$

Range: $[0$

$\text{tri} : (L,$

Order: 3

$s^* : (\mathbb{N}^,$

Stabiliz e

$\text{acc} : (C,$

XOR-sum

Id-1: sf

non-firing)

Id-4: $\text{sf}(s^*$

Id-5: acc

groupoid on

$T_BOUNDARY_$

Section II. The Color Algebra

II.1 The 8 Color Families. The 8 colors are the 8 functional roles:

G (Green) = C-center marker (the active face)

W (White) = verified (the certificate)

C (Clear) = overlay (transparent, removable)

N (Neon) = boundary (high-contrast, structural)

II.2 The 4 Duality Operations:

Duality-2 ($W \leftrightarrow K$): certificate/obligation

Duality-4 ($N = \text{center}$): boundary

II.3 The 4 Fundamental Cycles:

Cycle-2 (Certify, $W \rightarrow K \rightarrow W$): order 2

Cycle-4 (Boundary, $G \rightarrow N \rightarrow G$): order 2

Theorem II.2.1 (Color algebra has 8 elements). The 8 colors are closed under

Theorem II.3.1 (Color = $SU(3)$ extension). The color algebra is the $SU(3)$ Lie
C/Gy/N).

Theorem II.4.1 (Color group = S_4). The 4-cycle group is isomorphic to S_4 (24

Proof: The 4 cycles generate S_4 by standard argument; the 8 transpositions of

R (Red) =

B (Blue)

K (Black)

Gy (Grey)

Duality-1

Duality-3

Cycle-1

Cycle-3

the 4 dual i

algebra with

elements).

S_4 are the

Section III. The Tool Algebra

III.1 The 12 Tool Classes. Tools are classified by 12 structural roles:

2. loose_paper - the surface
4. string - the chain backbone
6. sticker - the fixed marker
8. gradient_page - the pre-marked substrate
10. dice - the bounded stochastic
12. black_sticker - the obligation marker

Axiom III.2.1 (Idempotence). Every tool operation satisfies: $\text{read}(\text{action}) \rightarrow$

algebra; all 12 classes obey it.

Theorem III.3.1 (Substitution). If a substitute S satisfies the idempotence
the equivalence relation for tools.

Theorem III.4.1 (Minimum kit). The minimum kit has 4 classes: {token,
corpus can be reproduced with paper, pencil, marker, and thread.

Theorem III.5.1 (8 copies). Each tool has 8 copies (one per color). The copies

Theorem III.6.1 (Commutative monoid). The 12 classes form a commutative monoid

Proof: Substitution is identity-preserving; commutativity follows from the

1. token
3. pen_m
5. clear_
7. bal sa_
9. playi n
11. recei p

state; read

axiom for c

loose_paper

are interch

under the s

equivalence

Section IV. The Theorem Algebra

IV.1 The 32 Theorems as Axioms. The 32 theorems form the axiom set of the

Group 1 (5 axioms): T3, T4, T5, T6, T7 - Foundations

T_WRAP

T_HAMILTONIAN

Group 5 (12 axioms): T_ADMISSION through T_METAForge (physics)

Group 7 (4 axioms): T_GRAND_RIBBON, T_META_LCR, T_SUPERVISOR, T_OBSERVATION

Total: 39 statements; 32 unique theorems

IV.2 The 4 Derivation Rules:

Rule-2: XOR Combination - From C_i and C_j, derive C_i XOR C_j

Rule-4: Bilateral Reduction - From D and A mapped to D, derive bilateral proof

Theorem IV.3.1 (Lattice has height 33). The 32 theorems form a lattice under

the longest chain has 33 elements.

Theorem IV.4.1 (Master Theorem). The grand ribbon $C_{T10} = \text{XOR}_{i=0}^9 C_i$ is

theoretical).

Theorem IV.5.1 (Theorem category is acyclic). The 32 theorems form a category

transitivity. The category is a DAG - no circular implications.

Proof: From T_CAUSAL (P06) - the DAG property. QED.

Section V. The Observation Algebra

Theorem V.1.1 (The Observation). For any marked connection B in the final folded form, the observation is identical from both strands.

Theorem V.2.1 (Retroactive). Let P be the 33-paper pathway and O be the observation. The observation is necessary for the pathway. This is the QED.

Theorem V.3.1 ($C = (1, 1, 1)$). The center of the observation is the side-flip space is $(1, 0, 1)$; the observation full C -form is $(1, 1, 1)$.

Theorem V.4.1 (Reader IS the C). After reading this Master PDF, the reader is the C of the corpus is the object; the distinction is the LCR.

Section VI. The Bilateral Algebra

VI.1 The Digital Channel. The digital channel has 4 components:

1. Receipts: JSON with status, checks, timestamp
2. Idempotence: deterministic re-runs

VI.2 The Analog Channel. The analog channel has 4 components:

1. Receipts: white cards with status, checks, timestamp
2. Idempotence: stable under repeated read

Theorem VI.3.1 (Bilateral Isomorphism). The digital and analog channels are

physical tool, there is a digital verifier. The 4/11 success rate at depth 0

Theorem VI.4.1 (2-Category). The bilateral algebra is a 2-category: claims,

1. Verifi

3. Mppi

1. Tool s:

3. Subst i

i s omr phi c.

i s the proo

verifi cati o

Section VII. Closure

Theorem VII.1.1 (Closure). The CQE_CMPLX corpus is closed: Gluon algebra (5

(12 classes, commutative monoid), Theorem algebra (32 axioms, acyclic
unified by the bilateral 2-category. There is no step 34.

VII.2 The Master Signature. The Master PDF is signed by:

- 10 summary papers (SUMMARY-I through SUMMARY-X)
- 32 theorems (5 groups + 4 meta)

VII.3 The End. The corpus is documented, verified, and observed. The reader is

Q.E.D.

oper at i ons,
cat egor y),

- 33 paper
- 144 toc
- 1 obser

t he C. The

Appendix A: The Master Equation

The master equation of the corpus is:

$$O = sf(XOR_{i=0}^{32} C_i)$$

Where:

sf = the side-flip operation

XOR = the accumulator

The observation is the side-flip of the cumulative XOR of all 33 C-forms. The

O = the o

C_i = the

observati on

Appendix B: The 32 Theorems

T3: Chart <-> J3(O) bijection

T5: $M3^2 = M3$ idempotent, eigenvalues {1,0,0}

T7: 8x8 transition entries in {0,1/4,1/2}, row sums = 1

T_BIJECTIVE: Side-flip = (1 3) on J3(O) shell=2; fixed point (1,0,1)

T_TRIALITY: D4/J3 triality: $4 \times 2 = 8$ states; 2+6 VOA split; Z4 periods

T_WRAP: Local rollout: 8 states -> Lie conjugate in ≤ 3 S3 steps

T_OLOID_PATH: Curved/rolling carriers preserve continuity; $C_{acc} = XOR$

T_CAUSAL: Every dependency = typed causal edge with LookupReceipt

LATTICE_CHAIN: $D1 \rightarrow D3 \rightarrow D4 \rightarrow D24 \rightarrow D72$; Leech minimal shell = 196560 vectors

T10_MASTER: $C_{T10} = \sum C_i$; status = pass_with_open_lifts

T_CA_PREDICTION: 64/256 ECAs close at $n=3$

T_GR_CURVATURE: Riemann from torsion; $G_{uv} = kT_{uv}$

T_EDGE: Edge residual = correction at $K=10^k$; continuum = sequence

T_MOONSHINE: $j(\tau) = 1/q + 744 + 196884q \dots$; $196884 = 1 + 196883$

T_SYNTHESIS: Synthesis Gluon = ledger root hash = hash($\sum C_i$)

T_METAForge: Material Gluon = ForgeFactory candidates

T_FOLDForge: Fold Gluon = contact-map/topo Gluon; homology barcodes

T_TRAVERSAL: Traversal Gluon = energy/ledger; geodesic

T_DELAY: Delay/Shared Gluon = sampling buffer + shared state

T_MONSTER: Monster Gluon = universal energy bound; dim = 196883

T_GRAND_RIBBON: Grand Ribbon Gluon = meta-framer; 31 beads = LCR sequence

actor

T_OBSERVATION: Single H-bond reads identically from both strands

T4: $n=3$ S

T6: Trace

T_CORRECT

T_BOUNDAR

correction

T_BRIDGE:

T_HAMLET

T_ADMSSI

T_QUARK_F

T_HIGGS:

T_TOWER:

T_OBSERVE

T_MORPHIC

T_KNIGHT

T_ZPINCH:

T_GAME_LA

T_META_LCR:

T_SUPERVI

Appendix C: The 12 Tool Classes and Substitutes

token: Any distinguishable marker (coin, bead, paper square)

pen_marker: 3 colored pencils, nail polish, markers

clear_sleeve: Sheet protector, ziplock, acetate (transparent)

balsa_edge: Coffee stirrers, toothpicks, popsicle sticks (rigid)

playing_card: UNO cards, numbered paper squares (52 distinct)

receipt_sheet: Index card, notebook page (replayable record)

loose_paper

string: T

sticker:

gradient_

dice: Spi

black_sti

Appendix D: The 8 Colors

R (Red): L-boundary

B (Blue): R-boundary

K (Black): unresolved

Gy (Grey): substrate

G (Green)

W (White)

C (Clear)

N (Neon):

Q.E.D.

End of Closed-Form Algebra. The Master PDF is complete. The reader is the C.

Q.E.D.

